Kielce University of Technology The Faculty of Electrical Engineering, Automatics and Computer Science			
FINAL report			
Course:	Object Oriented Programming 2		
Topic:	CAR RENTAL MANAGEMENT (BobrRent)		
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1. Executive Summary

BóbrRent is a robust, desktop-based car rental management solution. It utilizes a **Client-Server** model to provide real-time interaction between the user interface and the database. The system allows users to create accounts, browse a live fleet of vehicles, and manage rental transactions with automated price calculation and status updates.



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2. System Architecture

The application is structured into three distinct layers to ensure modularity and scalability:

- **Presentation Layer (Frontend):** Developed using **JavaFX** and **FXML**. It features a modern, card-based UI for car browsing and specialized controllers for handling user input.
- **Application Layer (Middleware):** A multi-threaded **Java Server** (BobrServer) that acts as a bridge. It listens on port 1234 for incoming socket requests and processes them via a custom communication protocol.
- **Data Layer (Backend):** An **Oracle XE Database** that stores all persistent information. It ensures data integrity through relational constraints and handles complex transactions using SQL.

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3. Database Design (Oracle SQL)

The database schema was refined to resolve initial identifier errors (such as ORA-00904). The final structure utilizes three primary tables:

3.1 Table Structures

- **CUSTOMERS:** Stores user credentials and profile details (Phone, Password, First/Last Name, Gender, Birthday).
- **CAR_INFO:** Manages the vehicle fleet status (Brand, Model, Price Per Day, Status).

- **RENTALS:** Records every successful booking transaction. This table was specifically created to handle the link between users and cars.

3.2 SQL Implementation

SQL

```
CREATE TABLE RENTALS (
   RENTAL_ID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
   CUSTOMER_PHONE VARCHAR2(20),
   BRAND VARCHAR2(50),
   MODEL VARCHAR2(50),
   START_DATE DATE,
   END_DATE DATE,
   TOTAL_PRICE NUMBER(10, 2),
   CONSTRAINT fk_rent_cust FOREIGN KEY (CUSTOMER_PHONE) REFERENCES
CUSTOMERS(PHONE)
);
```

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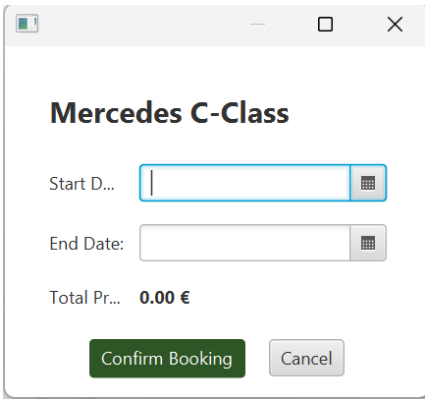
4. Key Functional Components

4.1 Session-Based Authentication

To provide personalized services, the system implements a UserSession utility. When a user logs in successfully, their unique phone identifier is stored in the application memory. This allows the **My Bookings** and **Profile** pages to display data relevant only to that specific user.

4.2 Automated Booking Workflow

1. **Selection:** User selects a car from the dynamic catalog.
2. **Calculation:** The BookingController calculates costs based on ChronoUnit.DAYS between selected dates.
3. **Transaction:** Upon confirmation, the server executes an **Atomic Transaction**:
    - Inserts rental details into the RENTALS table.
    - Updates the car's status to 'rented' in CAR_INFO.



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5. Problem Solving & Debugging

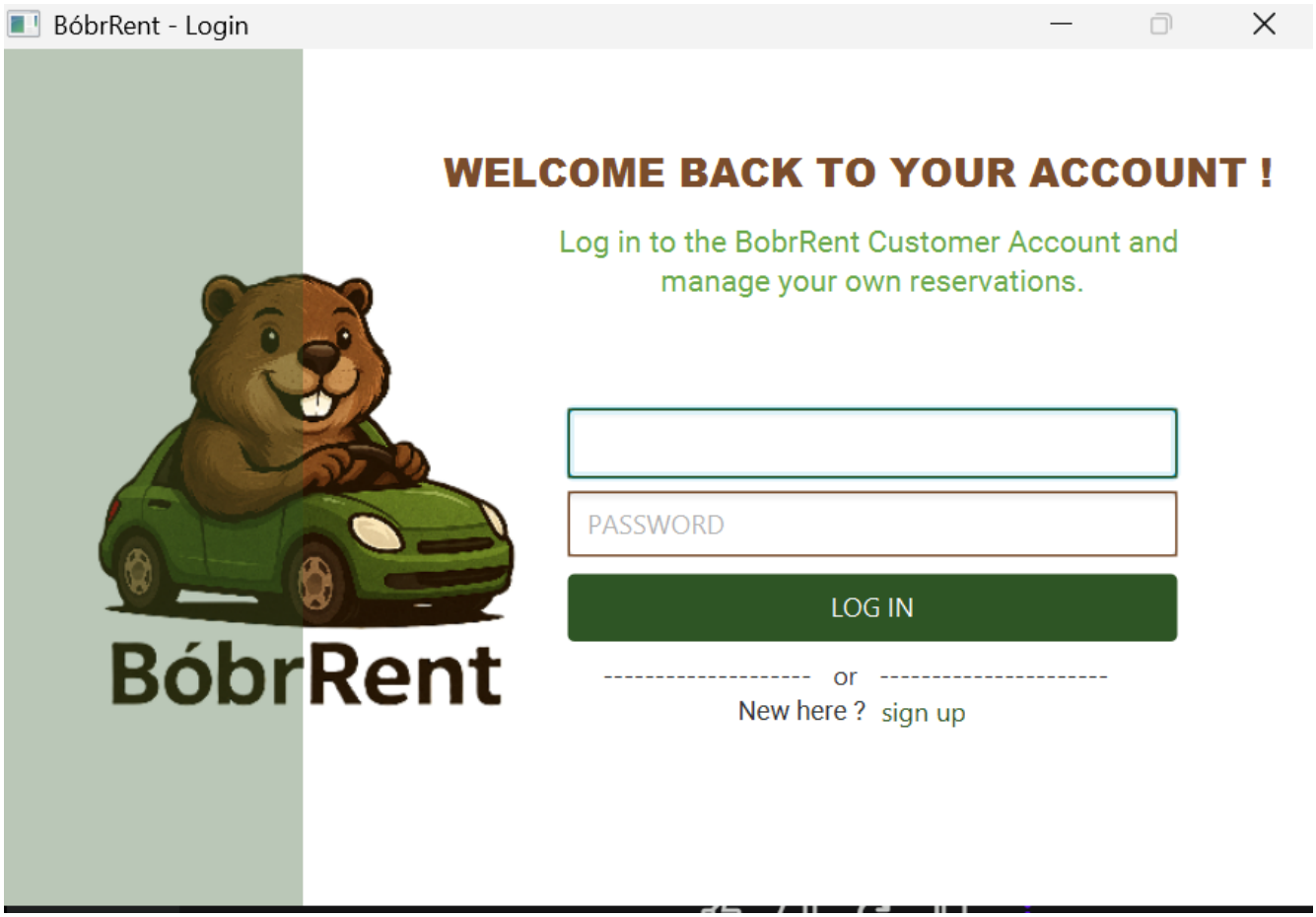
During development, several critical technical hurdles were addressed:

- **Syntax Correction:** Resolved the ORA-00904: "MODEL": invalid identifier error by ensuring the RENTALS table schema perfectly matched the Java server's SQL queries.
- **Resource Management:** Fixed the Location is not set crash in HomeController (line 42) by validating the FXML resource paths in the project directory.
- **Session Integrity:** Fixed a "Silent Failure" where bookings weren't saving by adding the missing UserSession.setPhone() call to the LoginController.

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6. Implementation Gallery

Detailed screenshots of the final working application.



BóbrRent - Login



BóbrRent

WE CAN'T WAIT TO WELCOME YOU AS PART OF OUT COMMINITY !

Sign up

FIRST NAME

LAST NAME

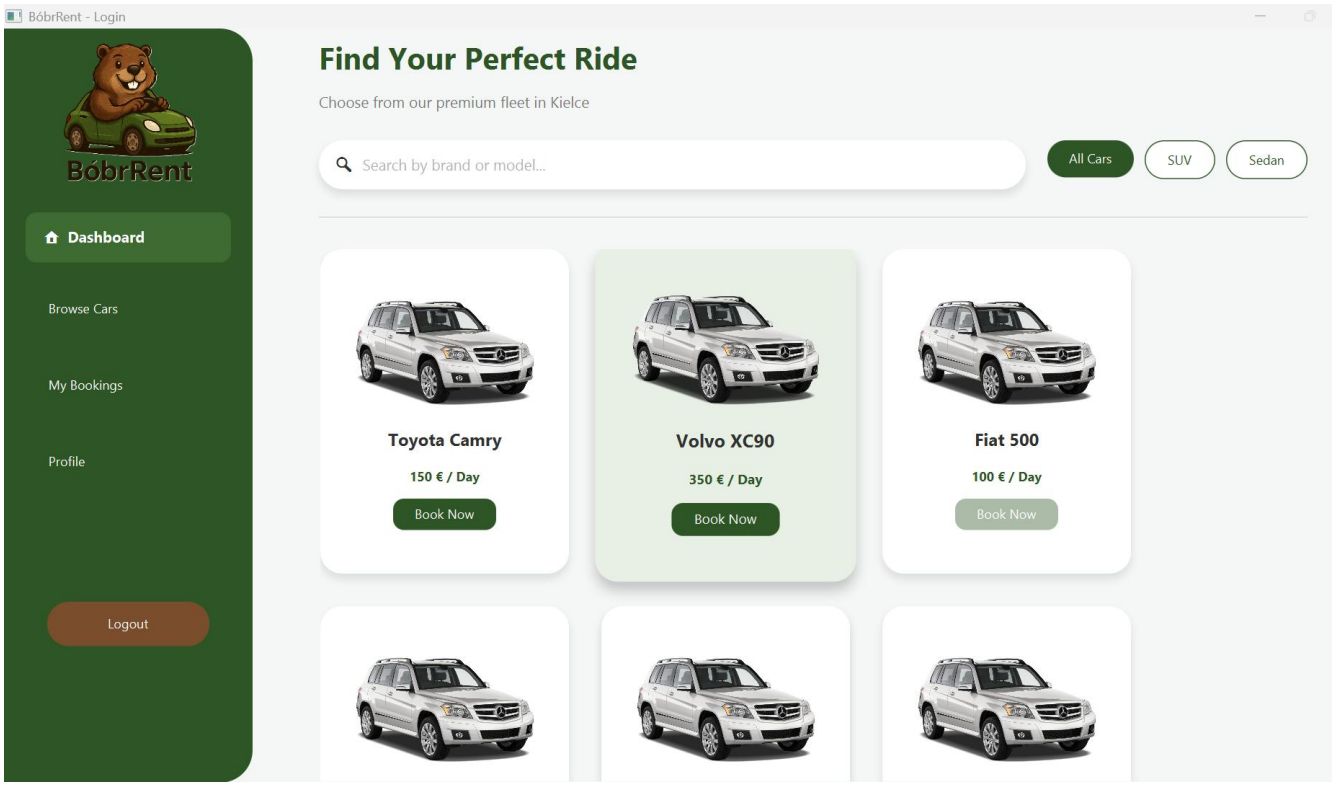
DATE OF BIRTH

PASSWORD

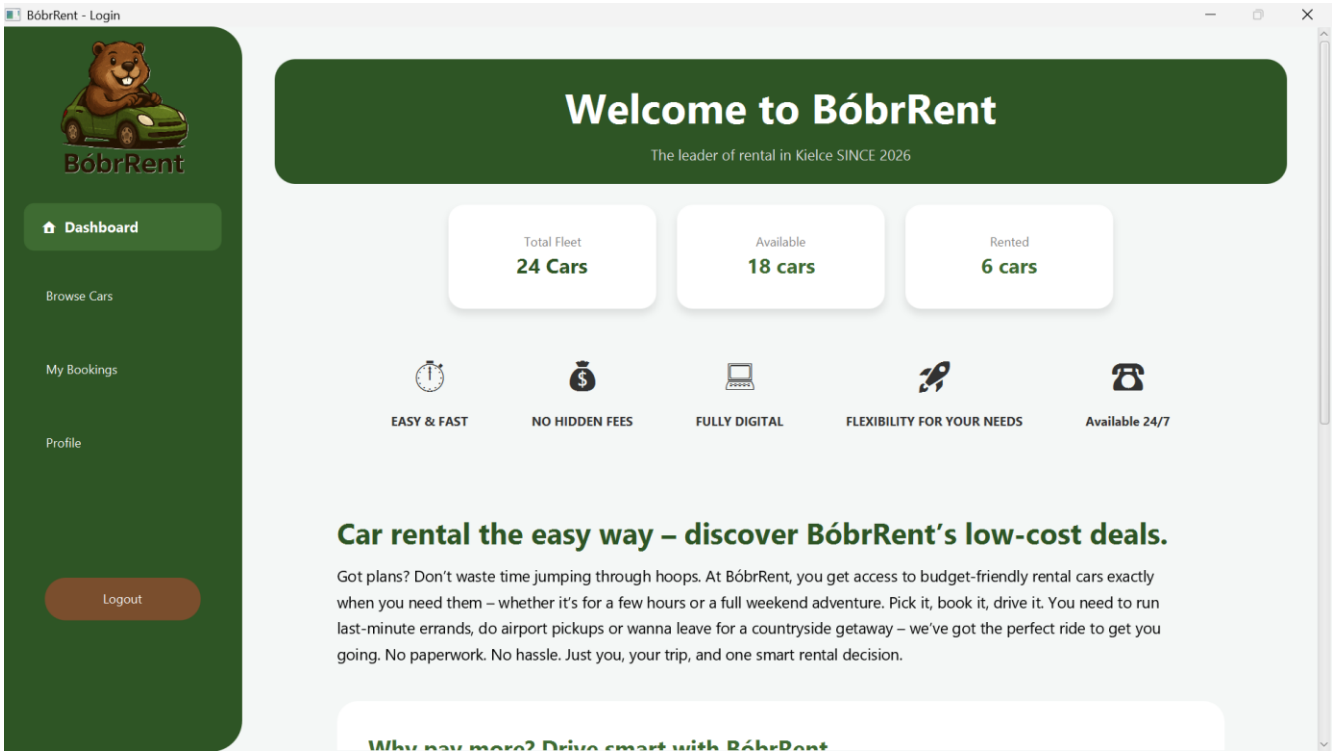
SIGN UP

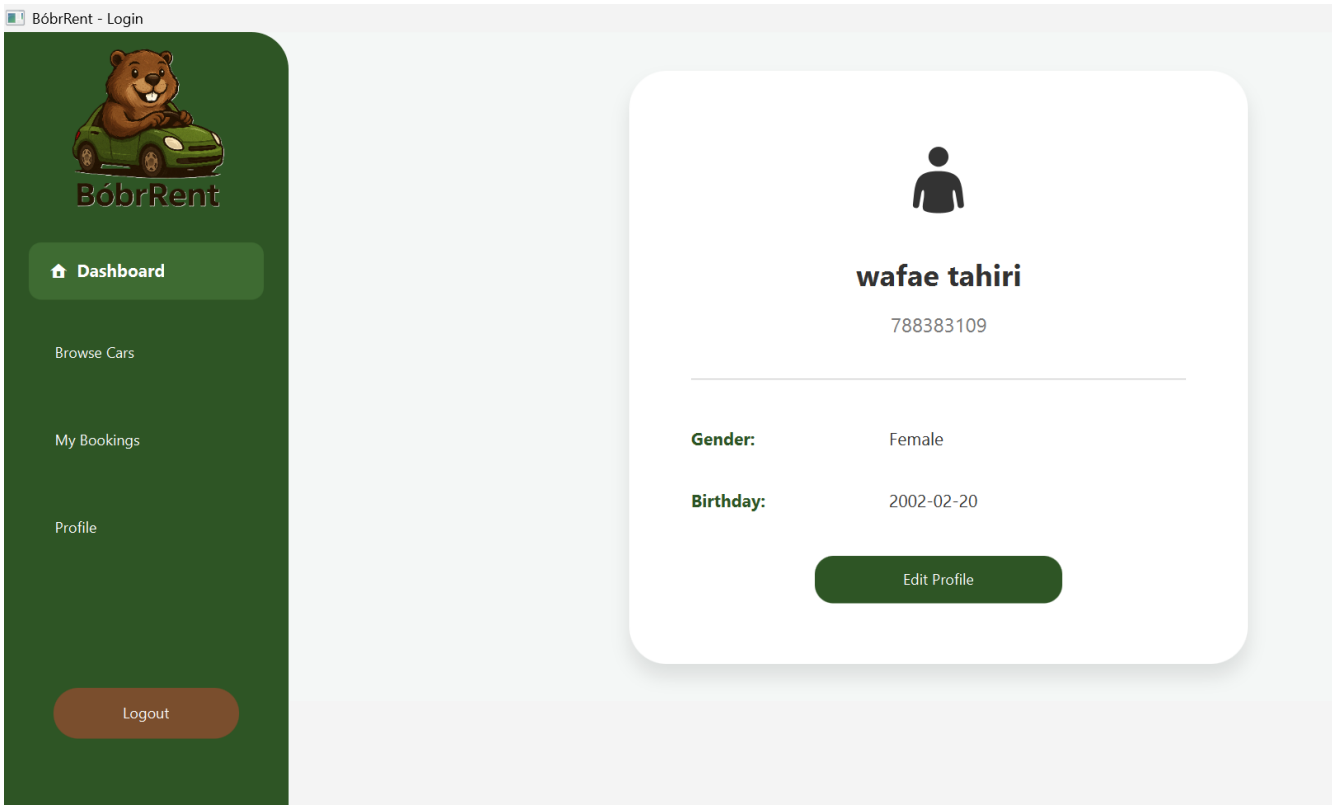
---------- or ----------

You already have an account? Login



Profile Details]





7. Conclusion

BóbrRent successfully integrates Java's object-oriented strengths with Oracle's data management capabilities. The final product is a reliable, user-friendly car rental application that maintains a high standard of data accuracy and transaction security.